Healthcare BI & Data Warehousing Project

Interactive Healthcare Performance Dashboard

Power BI | Power Query | DAX | Star Schema

Vanessa Negreiros
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Dataset Description

The dataset used in this project consists of eight healthcare-related tables, including one fact table (Fact Encounter) and seven dimension tables (Calendar, Patient, Provider, Facility, Diagnosis, Procedure, and Payer). The dataset contains information about patient encounters, demographic characteristics, diagnoses, medical procedures, healthcare providers, facilities, insurance payers, costs, charges, and admission dates.

The objective of this project was to transform the raw data into a structured data warehouse model, develop DAX measures, and build interactive dashboards to support descriptive business analysis and decision-making.

ETL Process

The ETL process was performed in Power Query and included data import, profiling, transformation, and validation. All eight CSV files were imported and renamed to improve readability and maintain a consistent naming convention throughout the project.

A comprehensive data profiling process was performed on every table to verify missing values, duplicate records, empty strings, data types, and data consistency. The analysis confirmed that the dataset was already clean and did not require extensive data cleansing before modeling.

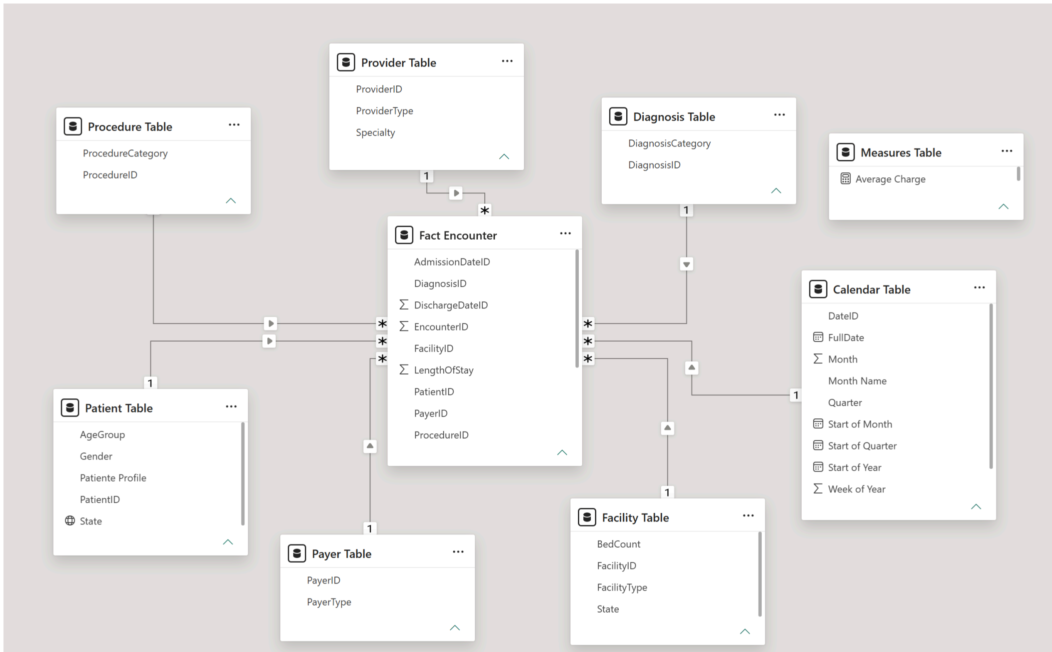
Table 1. Summary of ETL Transformations

Step	Description
Import	Imported eight CSV tables into Power BI.
Rename	Renamed all tables using descriptive names.
Data Profiling	Verified missing values, duplicates, empty strings, data types, and inconsistencies.
Calendar Table	Added Start of Year, Start of Quarter, Start of Month, Month Name, and Week of Year columns.
Patient Table	Created the Patient Profile column by combining Gender, Age Group, and State.
Fact Encounter	Created the Readmission Category column and converted Total Charges and Total Cost to the Fixed Decimal Number data type.

The completed ETL process produced a clean and standardized dataset suitable for dimensional modeling and business intelligence analysis.

Data Model and Relationships

Figure 1. Star Schema Data Model



The data model was designed following the Star Schema architecture, which is widely used in data warehousing to improve query performance and simplify business analysis. The Fact Encounter table was defined as the central fact table, while the Calendar, Patient, Provider, Facility, Diagnosis, Procedure, and Payer tables were modeled as dimension tables.

Each dimension table is connected to the fact table through a one-to-many relationship using the corresponding primary and foreign keys. All relationships are active and use a single filter direction from the dimension tables to the fact table, ensuring consistent filter propagation throughout the model.

To validate the model, a matrix visual was created using fields from multiple dimension tables together with measures from the fact table. The successful visualization confirmed that the relationships were configured correctly and that the Star Schema model functioned as expected.

DAX Measures

To support business analysis, several DAX measures were created using different categories of functions, including mathematical, statistical, counting, logical, CALCULATE, and Time Intelligence functions. These measures provide the key performance indicators displayed in the dashboards and enable interactive analysis under different filter contexts.

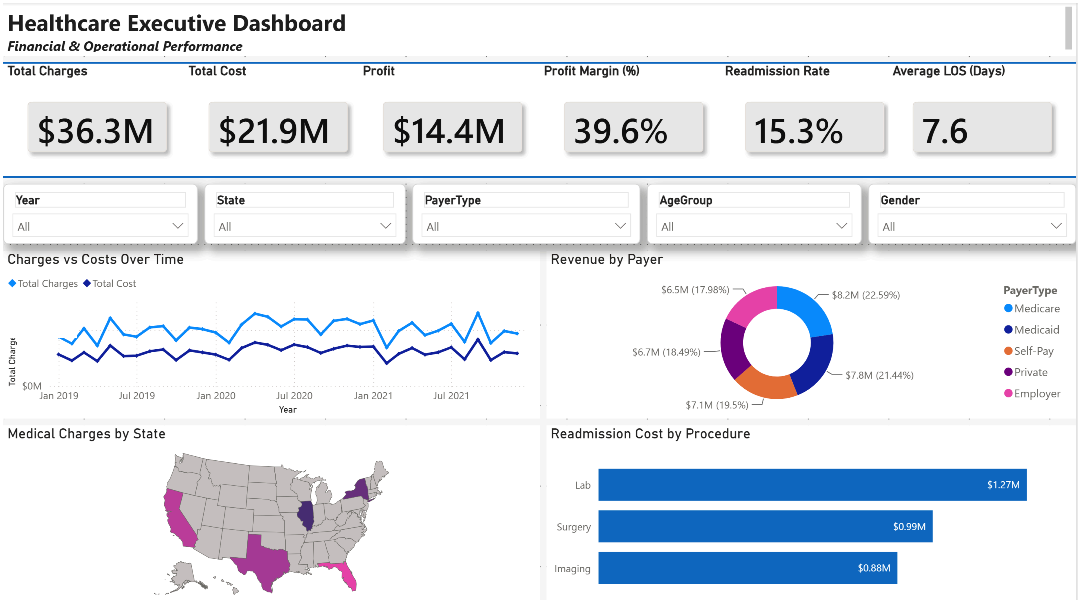
Table 2. DAX Measures

Measure	Category	Purpose
Total Charges	Math	Calculates the total amount charged to patients.
Total Cost	Math	Calculates the total hospital cost.
Profit	Math	Calculates the difference between total charges and total costs.
Profit Margin (%)	Math	Calculates the percentage profit margin.
Avg Charges	Statistical	Calculates the average charge per encounter.
Avg Costs	Statistical	Calculates the average cost per encounter.
Avg Length of Stay	Statistical	Calculates the average length of stay per encounter.
Total Encounters	Counting	Counts the total number of patient encounters.
Distinct Patients	Counting	Counts the number of unique patients.
Readmission Rate	Logical	Calculates the percentage of encounters resulting in readmission.
Readmitted Total Cost	CALCULATE	Calculates the total cost associated with readmitted patients only.
Charges YTD	Time Intelligence	Calculates cumulative total charges from the beginning of the year.
Previous Year Charges	Time Intelligence	Returns total charges for the corresponding period in the previous year.
YOY Growth (%)	Time Intelligence	Calculates the year-over-year growth rate of total charges.

Skills Demonstrated

- Data modeling and Star Schema design
- ETL and data profiling using Power Query
- DAX measures and Time Intelligence
- KPI development and financial analysis
- Interactive dashboard development
- Data visualization and business analysis

Executive Dashboard



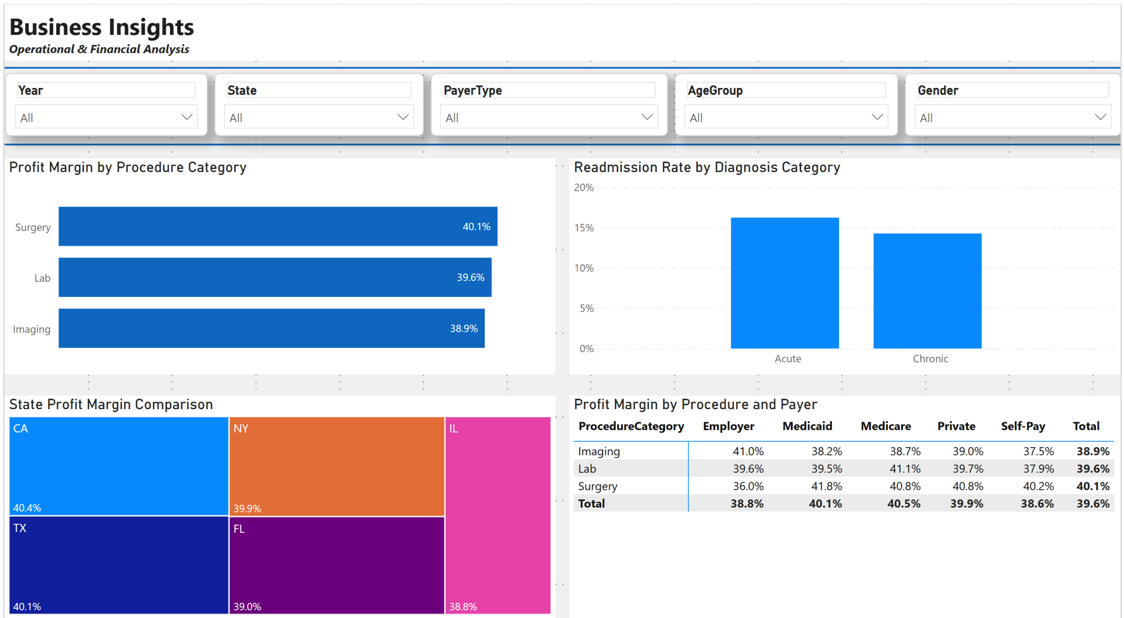
The Executive Dashboard provides a high-level overview of hospital performance by combining financial, operational, and geographical indicators in a single interactive report. Users can filter the analysis by Year, State, Payer Type, Age Group, and Gender, allowing dynamic exploration of the data from multiple perspectives.

The dashboard includes key performance indicators for Total Charges, Total Cost, Profit, Profit Margin, Readmission Rate, and Average Length of Stay. Additional visualizations summarize financial trends, revenue distribution by payer type, geographic distribution of charges, and readmission costs by procedure category.

Key Insights

- Strong overall financial performance: Total charges reached $36.3M compared with $21.9M in total costs, resulting in $14.4M in profit and an overall 39.6% profit margin across the analyzed period.
- Balanced revenue distribution: Revenue was distributed across multiple payer types, with no single payer accounting for a dominant share of total revenue. This indicates a relatively diversified payer mix.
- Readmission and operational performance: The overall readmission rate was 15.3%, while the average length of stay was 7.6 days. Readmitted costs also varied across procedure categories, highlighting areas that may warrant further operational investigation.

Business Insights



The Business Insights dashboard focuses on answering key business questions related to profitability and operational performance. Rather than providing only descriptive information, the dashboard was designed to support data-driven decision-making by exploring the following questions:

- Which procedure category has the highest profit margin?
- Which diagnosis category has the highest readmission rate?
- Which state has the highest profit margin?
- Which payer provides the highest profit margin across different procedure categories?

The visualizations allow users to compare financial performance across procedures, evaluate readmission patterns by diagnosis, identify geographical differences in profitability, and analyze the relationship between payer type and procedure profitability.

Key Insights

• Surgery achieved the highest profit margin among the analyzed procedure categories, at 40.1%.

• Profitability varied across states and payer types, with differences providing opportunities for more targeted financial and operational analysis.

Limitations and Optimization

Limitations: The dataset contains only 1,500 encounter records over a three-year period and represents a simplified healthcare dataset. The limited number of observations and available variables restrict more detailed trend analysis and advanced predictive modeling.

Optimization: Model performance was improved by implementing a Star Schema, optimizing data types, validating unique primary keys, and configuring active one-to-many relationships with a single filter direction. In addition, Power Query was used for data transformations whenever possible, and all DAX measures were organized into a dedicated Measures Table to improve model organization and maintenance.

Conclusion

This project demonstrates an end-to-end Power BI workflow, from data profiling and ETL through Star Schema modeling, DAX measure development, and interactive dashboard design. The analysis combines technical data skills with business-oriented analysis to evaluate healthcare financial and operational performance. The resulting dashboards provide a structured view of profitability, payer mix, readmissions, procedures, and geographic performance, supporting data-driven decision-making while highlighting opportunities for further analysis with larger and more detailed datasets.